

Deep Learning

Instructor Name: Mamoon Akbar

TA Name (if any):

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Office Location/Number: New building room 50

Office Hours:

Course Information Program: BSDS (7A,7B)

Credit Hours: 3

Core Prerequisites (if any):

Course Website (if any) : Google classroom

Class Meeting Time:

Textbook(s) /Supplementary Readings:

There is no standard one "textbook" for this course. The following book will be used as a primary text to guide some of the discussions, but it will be heavily supplemented with lecture notes and reading assignments from other sources.

- Neural Networks and Deep Learning by Michael Nielsen
<http://neuralnetworksanddeeplearning.com/>
- Deep Learning, An MIT Press book, by Ian Goodfellow and Yoshua Bengio and Aaron Courville <http://www.deeplearningbook.org/>

Additional references and books related to the course:

- <https://www.deeplearningbook.org/>
- Book: <http://neuralnetworksanddeeplearning.com/>
- Deep Learning with Python, by Francois Challet, Manning Publications
- Introduction to Machine Learning by Ethem Alpaydm (latest edition)
- Machine Learning: A Probabilistic Perspective by Kevin P. Murphy

Tentative Course Outline

Week	lecture	Topics
1	introduction	• Overview – Brain – Neuron - Hubel & Wiesel, 1959 • What is Learning? • What is Machine Learning • Historical Context Shallow • Feature Learning • What is Classification? • Shallow vs. Hierarchical vs. Deep Features (ML vs. DL) • Linear regression

		• Logistic Regression
2	Neural Network	• Gradient Descent • Perceptron • Forward and Back propagation
3	Encoder , Decoder	• Gated Recurrent unit • Bidirectional RNN • CHATGPT • Transformer • Visual Transformer
4	Autoencoders	• Autoencoder Architecture • Sparsity in Autoencoders • Stacked Autoencoders
5	Mid 1	
6	Segmentation	• Image Segmentation (Unet), • Instance Segmentation • Image Retrieval using Unsupervised/Semi-supervised Learning
7-8	GANS	• GANS • Adversarial Learning • Adversarial Attacks • Adversarial AAE • Discriminative AAE • Cyclic GANS – CT+X-Ray • Cross-View Generation – Remote Sensing
9	Geometric Deep Learning	• Applying Deep Learning on 3D Data • Graph-based edge convolution • Learnable local (non-linear) operator
10	Mid 2	
11	Geometric Deep Learning	• Dynamic Graph Neural Network • Different Formulation of non-Euclidean • CNNs • Reimannian Geometry Calculus on Manifolds

12-13	Deep Reinforcement Learning	Markov Decision Process Q-Learning Tool box The Bellman Equation Deep Q-Learning Policy Gradients
14-15	CNN & Transfer learning	Research papers AlexNet, VGG , ResNet, GoogleLeNet etc
Final		

(Tentative) Grading Criteria

Quizzes 10%

Assignments/Homeworks/Project 10 - 20%

Midterms 30%

Final Exam 40%

Total: 100